

torch.autograd.function.FunctionCtx.set

<https://pytorch.org/docs/stable/generated/torch.autograd.function.FunctionCtx>

Description:

Set whether to materialize grad tensors. Default is True. This should be called only from either the `setup_context()` or `forward()` methods. If True, undefined grad tensors will be expanded to tensors full of zeros prior to calling the `backward()` and `jvp()` methods.

Function Signature

```
FunctionCtx.set_materialize_grads(value)[source]#
```

Code Examples

```

>>> class SimpleFunc(Function):
>>>     @staticmethod
>>>     def forward(ctx, x):
>>>         return x.clone(), x.clone()
>>>
>>>     @staticmethod
>>>     @once_differentiable
>>>     def backward(ctx, g1, g2):
>>>         return g1 + g2  # No check for None necessary
>>>
>>> # We modify SimpleFunc to handle non-materialized grad
>>> outputs
>>> class Func(Function):
>>>     @staticmethod
>>>     def forward(ctx, x):
>>>         ctx.set_materialize_grads(False)
>>>         ctx.save_for_backward(x)
>>>         return x.clone(), x.clone()
>>>
>>>     @staticmethod
>>>     @once_differentiable
>>>     def backward(ctx, g1, g2):
>>>         x, = ctx.saved_tensors
>>>         grad_input = torch.zeros_like(x)
>>>         if g1 is not None:  # We must check for None now
>>>             grad_input += g1
>>>         if g2 is not None:
>>>             grad_input += g2
>>>         return grad_input
>>>
>>> a = torch.tensor(1., requires_grad=True)
>>> b, _ = Func.apply(a)  # induces g2 to be undefined

```

Detailed Documentation

Documentation

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